SENTINEL-GNSS — Individual Poster

EKF Implementation, Real-Time Dashboard, and Live System Integration

Mustapha Ibrahim (LS2525238)   Systems Engineer

Beihang University H3I · 2025

My Role

I built the systems layer of SENTINEL-GNSS:

- 9-state EKF skeleton with adaptive R formula
- Simulated blockage EKF validation
- All five dashboard panels (Plotly Dash prototype)
- 1 Hz real-time data playback integration
- Room 3058 Septentrio receiver connection
- Live NMEA/UBX parsing and feature streaming
- End-to-end latency measurement: **82 ms** (target: <100 ms)
- Paper Section 6: System Demonstration

9-State EKF I Implemented

State vector:

$$\mathbf{x} = [x, y, v_x, v_y, \psi, b, b_{a_x}, b_{a_y}, z]^\top \in \mathbb{R}^9$$

Adaptive R formula:

$$R = R_{\text{base}} \times (1 + 10 \cdot P_{\text{DEGRADED}})$$

When $P = 0$: standard EKF ($R = R_{\text{base}}$)
When $P = 1$: GPS weight $\approx 8\,\%$ of standard

Simulated validation:
Fed 60-second blockage sequence; GNSS weight plot showed expected drop when P_{DEGRADED} rose to 1.0

Dashboard Panel 1 — Map

Interactive Leaflet/Folium map with:

- Full GPS track coloured **CLEAN** / **WARNING** / **DEGRADED** by predicted class
- Moving vehicle marker advancing at 1 Hz
- Predicted-quality halo around current position

Dashboard Panel 2 — Signal Quality

Three scrolling Plotly time-series at 1 Hz:

- Mean C/N_0 with 30 / 35 dB-Hz threshold lines
- SV count with minimum-4 alert line
- PDOP with 3.0 alert threshold

Dashboard Panel 3 — Prediction Gauges

Three semicircular probability gauges (+5s, +15s, +30s):

- Gauge sweep shows P(CLEAN)/P(WARNING)/P(DEGI
- Status banner: **NOMINAL** / **CAUTION** / **ALERT**
- Visual "fuel-gauge" representation for non-technical users

Dashboard Panel 4 — Sky Plot

Azimuth–elevation circular plot (one dot per satellite):

- **Green**: $C/N_0 > 40$ dB-Hz (healthy)
- **Amber**: C/N_0 30–40 dB-Hz (marginal)
- **Red**: $C/N_0 < 30$ dB-Hz (degraded)

During the live campus test: three satellites switched green→red as we entered the building shadow, immediately before the ALERT fired.

Dashboard Panel 5 — Fusion Status

Stacked horizontal bar showing real-time GNSS/IMU weighting:

$$\text{GNSS\%} = 100\cdot(1-P_{\text{DEGRADED}}), \quad \text{IMU\%} = 100\cdot P_{\text{DEGRADED}}$$

Makes the prediction tangible: bar shifts visually from green-GNSS to blue-IMU as the vehicle enters the building shadow.

Live Receiver Integration

Connected Septentrio Mosaic-X5C (Room 3058) via USB/serial:

```
ser = serial.Serial('COM8', 115200)
```

Parsed: $GPGGA, $GPRMC, $GPGSV sentences.
Extracted: lat/lon/alt, SV count, HDOP, per-SV C/N_0.

Live feature extraction verified:

- 1 Hz epoch rate, <2 ms jitter
- C/N_0 values consistent with historical RTKLIB results
- Feature extraction <15 ms/epoch

End-to-End Latency

Stage	Time (ms)
Feature extraction	12
SENTINEL inference	0.039
Dash callback + serialise	48
Browser render	22
Total	**82 ms**

Target: <100 ms. Achieved: 82 ms.

Dominant cost: Plotly serialisation (48 ms). Resolved in the production Next.js version using WebSocket push.

Key Lesson

Sky plot is the most compelling demonstration tool. No table of RMSE numbers communicates the physics of satellite blockage as clearly as watching satellite dots turn red in real time. For stakeholder communication, one animated sky plot is worth 10 charts.